

3

Graphs of Polytopes

The vertices and edges of a d -polytope P form an undirected graph $G(P)$ that encodes a lot, but not everything, about the combinatorial structure of the polytope.

In this lecture we discuss three fundamental topics about graphs of polytopes: the monotone Hirsch conjecture (for which we prove validity for $0/1$ -polytopes, and Kalai's recent bound for the general case), Kalai's reconstruction of simple polytopes, and Balinski's d -connectivity theorem.

Before we look into this, we will establish two technical tools: the power and the glory of half-sentences like “let L be a line in general position,” and the (geometric version of the) simplex algorithm for linear programming, which is the most fundamental search technique on polytopes.

3.1 Lines and Linear Functions in General Position

We start with a short discussion of the concept of “general position with respect to P ” — this will yield a few useful tools for proofs in subsequent sections. To illustrate this, we will sketch a quick-and-dirty geometric version of linear programming in the next section.

With all the work we did in Lectures 1 and 2, we can now assume that we are dealing with a d -dimensional polytope P in $\mathbb{R}^d$, in which $\mathbf{0}$ is an interior point. We alternate freely between representations in terms of vertices, $P = \text{conv}(V)$, and in terms of facets, $P = P(A, \mathbf{1})$. Here $A \in \mathbb{R}^{n \times d}$ is a matrix that is considered as a set of rows, $A = \{\mathbf{a}_1, \dots, \mathbf{a}_n\}$.

We also need the hyperplanes in $\mathbb{R}^d$ determined by the facets of P , for which we introduce the notation

$$H_i := \{\mathbf{x} \in \mathbb{R}^d : \mathbf{a}_i \mathbf{x} = 1\}$$

— thus the facets of P are given by $F_i = H_i \cap P$. Each of these hyperplanes H_i determines two halfspaces, where

$$H_i^- := \{\mathbf{x} \in \mathbb{R}^d : \mathbf{a}_i \mathbf{x} \leq 1\}$$

denotes the closed halfspace that contains P , and H_i^+ denotes the other closed halfspace. In particular,

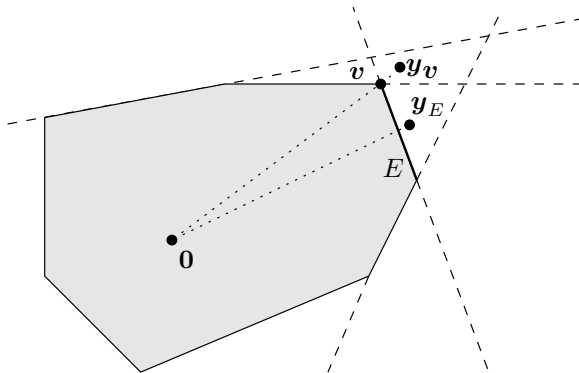
$$P = H_1^- \cap H_2^- \cap \dots \cap H_n^-.$$

As a start, for every face $F \in L(P)$ we know how to find a point $\mathbf{x}_F \in \text{relint}(F)$ in the relative interior of F : for example, we can take the barycenter of the set of vertices of F .

A very useful thing to get is a point outside P but very close to F — this is used for *stellar subdivisions* (see Exercise 3.0), Schlegel diagrams (see Lecture 5), and many other constructions. In particular, $\mathbf{x}_F$ does not lie on any of the hyperplanes H_i ; it lies “in general position” with respect to the hyperplane arrangement $(\mathbb{R}^d, \{H_1, \dots, H_n\})$ determined by P .

For this, let F be a proper face of P , and define $\mathbf{y} \in \mathbb{R}^d$ to be a *point beyond* F if $\mathbf{y}$ and $\mathbf{0} \in \text{int}(P)$ lie on different sides of H_i for every facet-defining hyperplane H_i that contains F , but on the same side of H_j for every facet-defining hyperplane H_j that does not contain F .

In other words, $\mathbf{y}$ lies beyond F if it satisfies $\mathbf{a}_i \mathbf{y} > 1$ for every inequality that is valid with equality for F , but $\mathbf{a}_j \mathbf{y} < 1$ for every other inequality. Our sketch shows a convex polygon (2-polytope) and indicates a vertex $\mathbf{v}$, a point $\mathbf{y}_v$ beyond that vertex, an adjacent edge E , and a point $\mathbf{y}_E$ beyond E .



How do we find $\mathbf{y}_F$? Well, we can take it on the ray that emanates from $\mathbf{0}$ and goes through a relative interior point $\mathbf{x}_F$. In fact, we can take $\mathbf{y}_F := t\mathbf{x}_F$ for any $t > 1$ such that $\mathbf{a}_i(t\mathbf{x}_F) < 1$ whenever $0 < \mathbf{a}_i \mathbf{x}_F < 1$. Clearly this t can be found (explicitly).

The next object we sometimes need is a line “in general position.”

Definition 3.1. A line through $\mathbf{0} \in \text{int}(P)$ is *in general position* with respect to P if it is not parallel to any of the hyperplanes H_i and does not hit the intersection of any two of them.

If the line is written in the form $L(\mathbf{u}) = \{t\mathbf{u} : t \in \mathbb{R}\}$ (for some $\mathbf{u} \neq \mathbf{0}$), then general position means that $\mathbf{a}_i\mathbf{u} \neq 0$ and $\mathbf{a}_i\mathbf{u} \neq \mathbf{a}_j\mathbf{u}$ for all $1 \leq i, j \leq n$.

The following lemma shows that a direction vector for such a line can be found arbitrarily close to any given vector.

Lemma 3.2. Let $P = P(A, \mathbf{1})$, and let $\mathbf{u} \in \mathbb{R}^d \setminus \mathbf{0}$. If $\lambda > 0$ is small enough, then the line $L(\mathbf{u}^{(\lambda)})$ is in general position with respect to P , for

$$\mathbf{u}^{(\lambda)} := \mathbf{u} + \begin{pmatrix} \lambda \\ \lambda^2 \\ \vdots \\ \lambda^d \end{pmatrix}.$$

Proof. We use that $s_i(\lambda) := \mathbf{a}_i\mathbf{u}^{(\lambda)} = \sum_{k=1}^d a_{ik}(u_k + \lambda^k)$ is a nonvanishing polynomial in λ of degree at most d , which has at most d positive zeroes. The polynomials $s_i(\lambda)$ are distinct, since $\mathbf{a}_i \neq \mathbf{a}_j$ for $i \neq j$.

From this we get that $\mathbf{a}_i\mathbf{u}^{(\lambda)} \neq 0$ for all, except at most d , positive values of λ , and that $\mathbf{a}_i\mathbf{u}^{(\lambda)} \neq \mathbf{a}_j\mathbf{u}^{(\lambda)}$ for all positive values with not more than $\binom{n}{2}d$ exceptions. $\square$

A remark about *being explicit*: mathematicians might tend to use either topological arguments (“a finite set of hyperplanes is nowhere dense in $\mathbb{Q}^d$ ”) or unnecessary algebraic machinery (“let x_{ij} be a set of $d \cdot n$ independent transcendentals over the ground field”). For discrete problems like those posed by our polytope applications, this is unnecessary.

The construction of Lemma 3.2, which we use to find points, lines, etc., in general position, depends on perturbation by some $\lambda > 0$ that has to be chosen small enough. It is not hard to be even more precise and completely explicit: it is (in principle) easy to *compute* a bound λ_0 such that every λ with $0 < \lambda < \lambda_0$ is small enough. This is because for polynomials with rational coefficients, one can bound the positive zeroes away from 0. A good reference for the ideas used for such explicit bounding is Lovász’ lecture notes [368, Ch. 1].

(The small positive parameter λ that we need would usually be called ε , but that would make this look like a course in analysis. We’ll try to avoid this until Lecture 9, where we start to integrate over polytopes.)

The same method of proof as used in Lemma 3.2 also yields the existence of a hyperplane in general position, arbitrarily close to a given one. The way to get this is to perturb the coefficients of the linear function that defines the hyperplane. For this, we state the result, but skip the proof.

Definition 3.3. A linear function $\mathbf{c}\mathbf{x}$ on $\mathbb{R}^d$ is *in general position* (or *generic*) with respect to a polytope $P \subseteq \mathbb{R}^d$ if it separates the vertices of P , that is, if $\mathbf{c}\mathbf{v}_i \neq \mathbf{c}\mathbf{v}_j$ for any two distinct vertices $\mathbf{v}_i, \mathbf{v}_j$ of P .

Lemma 3.4. Let $P = P(A, \mathbf{1})$, and let $\mathbf{c} \in (\mathbb{R}^d)^* \setminus \mathbf{0}$. If $\lambda > 0$ is small enough, then the linear function $\mathbf{c}^{(\lambda)}\mathbf{x}$ is in general position with respect to P , for

$$\mathbf{c}^{(\lambda)} := \mathbf{c} + (\lambda, \lambda^2, \dots, \lambda^d). \quad \square$$

3.2 Directing the Edges (“Linear Programming for Geometers”)

Definition 3.5. Let P be a convex polytope. The vertices and the edges of P form an abstract, finite, undirected, simple graph, called the *graph of P* and denoted by $G(P)$.

For every face $F \in L(P)$, we denote by $G(F)$ the *induced subgraph* of $G(P)$ on the subset $\text{vert}(F) \subseteq \text{vert}(P)$ of the vertices of $G(P)$, that is, the graph of all vertices in F , and all edges of P between them. This coincides with the graph of F , if F is itself considered as a polytope.

(In this whole course we need very little graph theory, only some terminology. When in doubt, look it up in any graph theory book. For that purpose, even a mediocre book would do. As for good ones, we recommend Bondy & Murty [123], Bollobás [121], or Tutte [551].)

We will consider *orientations* of $G(P)$, which assign a *direction* to every edge. An orientation is *acyclic* if there is no directed cycle in it. This implies (because all our graphs are finite) that there is a *sink*: a vertex that does not have an edge directed away from it. (Proof: Start at any vertex, and keep on walking along directed edges until you close a directed cycle or get stuck in a sink.)

Linear programming is (in a geometer’s version) the task to find a point $\mathbf{x}_0 \in P$ that maximizes a linear function $\mathbf{c}\mathbf{x}$, that is, such that $\mathbf{c}\mathbf{x}_0 = \max\{\mathbf{c}\mathbf{x} : \mathbf{x} \in P\} =: c_0$. Now we easily see that the maximum is achieved in a vertex. In fact, $F_0 := \{\mathbf{x} \in P : \mathbf{c}\mathbf{x} = c_0\}$ is a face of P , and thus every vertex of F_0 maximizes $\mathbf{c}\mathbf{x}$.

